DISTRICT HORTICULTURAL SOIL & NUTRITION TESTING LAB, KOLAR

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA09	**Report Date:**	23-Oct-2026
Location / District:	Malur Taluk, Kolar District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Red Gravelly Loam (Alfisol)	**Target Agro-Zone:**	Karnataka - Kolar Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**152.0**	kg/ha	280.0 - 560.0	Low (Deficient)
Available Phosphorus	**P**	**68.0**	kg/ha	23.0 - 56.0	High (Fertilizer Build-up)
Available Potassium	**K**	**185.0**	kg/ha	140.0 - 330.0	Medium (Adequate)
Soil Reaction (pH)	**pH**	**6.90**	pH scale	6.5 - 7.5	Neutral (Optimal)
Electrical Conductivity	**EC**	**0.65**	dS/m	< 1.00	Normal (Safe)
Organic Carbon	**OC**	**0.44**	%	0.50 - 0.75	Low (Depleted)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 152.0 kg/ha [Low (Deficient)]

Available Phosphorus (P): 68.0 kg/ha [High (Fertilizer Build-up)]

Available Potassium (K): 185.0 kg/ha [Medium (Adequate)]

Soil Reaction (pH): 6.90 [Neutral (Optimal)]

Electrical Conductivity (EC): 0.65 dS/m [Normal (Safe)]

Organic Carbon (OC): 0.44 % [Low (Depleted)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: Horticultural zone soil exhibiting high residual phosphorus accumulation from prolonged intensive DAP applications, alongside depleted nitrogen and organic humus reserves.

General Nutrient Management Guidance: Omit phosphatic basal fertilizers (DAP/SSP) for one crop season to deplete surplus soil P. Supplement nitrogen and organic carbon using vermicompost @ 3 t/ha and bio-fertilizer Azotobacter. Ideal for tomato, capsicum, carrot, and mulberry.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*